

MAC

SIMD

SIMD/Vector Extensions

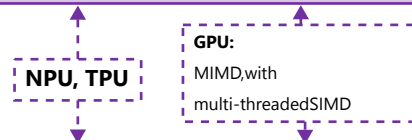
Subword-parallelism (**FPGA**)

Vektorrechner:

Pipelining of computation
Vector-register

Feldrechner (Array):

Multiple processing units
E.g. MAC : $A=A+B \cdot C$

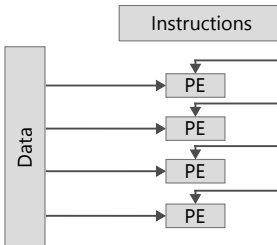


MIMD

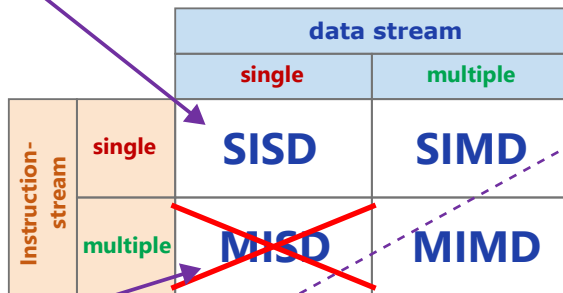
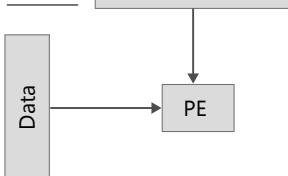
Multiprozessor-calculator

- Classification by memory access
- Classification by memory and address space structure

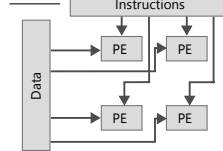
SIMD



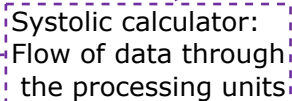
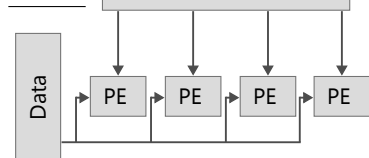
SISD



MIMD



MISD



Systolic calculator:
Flow of data through
the processing units

SISD

Von-Neumann architecture:

Shared memory for
operands and instructions;
shared memory processor
interface

Harvard architecture and

Harvard-Cache architecture:

Separate memory and
interface for instructions
and operands

MISD

no realized computers